

Unit III Question Bank

MCQs

1. What is the key idea behind Recurrent Neural Networks (RNNs)?

- A) They use only feedforward connections.
- B) They have an "internal state" updated as a sequence is processed.
- C) They are only used for static image classification.
- D) They cannot handle variable-length sequences.

Correct Answer: B

2. Which recurrence formula is typically used at every time step in an RNN?

- A) $h_t = f_W(h_{t-1}, x_t)$
- B) $h_t = f_W(x_t)$
- C) $h_t = f_W(h_{t+1}, x_t)$
- D) $h_t = f_W(h_{t-1})$

Correct Answer: A

3. In a "Vanilla RNN," what is the formula for the hidden state h_t ?

- A) $h_t = \sigma(W_{xh}x_t)$
- B) $h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t)$
- C) $h_t = \text{ReLU}(W_{hh}h_{t-1})$
- D) $h_t = W_{hy}h_t$

Correct Answer: B

4. What is another name for a "Vanilla RNN"?

- A) ResNet
- B) LSTM
- C) Elman RNN
- D) AlexNet

Correct Answer: C

5. Which architecture is suitable for "Image Captioning"?

- A) One to one
- B) One to many
- C) Many to one

D) Many to many

Correct Answer: B

6. Which architecture is suitable for "Action Prediction" from video frames?

A) Many to one

B) One to many

C) One to one

D) Many to many

Correct Answer: A

7. What does the "many-to-many" RNN architecture typically represent in video analysis?

A) Video classification on frame level

B) Single image classification

C) Encoding a whole sequence into one vector

D) Generating a single caption for a whole video

Correct Answer: A

8. Which of the following is NOT a design criterion for sequence modeling?

A) Handle variable-length sequences

B) Track long-term dependencies

C) Share parameters across the sequence

D) Use different weights for every time step

Correct Answer: D

9. In an RNN, what is reused at every time step?

A) The input vector x_t

B) The same set of parameters/weight matrix W

C) The output vector y_t

D) The loss function L

Correct Answer: B

10. What is the standard algorithm for training RNNs?

A) Standard backpropagation

B) Recirculation

C) Backpropagation through time (BPTT)

D) Contrastive divergence

Correct Answer: C

11. What is a major challenge when training Vanilla RNNs on long sequences?

A) They are too fast.

B) Vanishing or exploding gradients

C) They have too many parameters for short inputs.

D) They cannot process text data.

Correct Answer: B

12. What is "Truncated Backpropagation through time"?

A) Backpropagating through the whole sequence at once.

B) Training on only the first- and last-time step.

C) Running forward/backward through chunks of the sequence instead of the whole thing.

D) Stopping training after 10 epochs.

Correct Answer: C

13. Why do vanishing gradients occur in RNNs?

A) Multiplying many small numbers (gradients) together over many steps.

B) Using too many ReLU activations.

C) The weights are always set to zero.

D) The input is too large.

Correct Answer: A

14. Which technique is used to solve "Exploding Gradients"?

A) Weight decaying

B) Gradient clipping

C) Adding more layers

D) Using a smaller learning rate only

Correct Answer: B

15. Which activation function is a suggested solution to help prevent vanishing gradients?

A) Sigmoid

B) Tanh

C) ReLU

D) Linear

Correct Answer: C

16. How can weight initialization help with vanishing gradients?

A) Initialize weights to zero.

B) Initialize weights to the identity matrix.

C) Initialize weights to very large numbers.

D) Initialize weights to a constant negative value.

Correct Answer: B

17. What is the primary purpose of "Gates" in LSTM and GRU?

A) To increase the number of parameters.

B) To selectively add or remove information within each recurrent unit.

C) To make the model harder to train.

D) To prevent the model from using the hidden state.

Correct Answer: B

18. Which of the following is NOT a gate in an LSTM cell?

A) Forget gate

B) Input gate

C) Reset gate

D) Output gate

Correct Answer: C

19. What does the "Forget Gate" in an LSTM do?

A) It resets the entire network.

B) It decides what information to ignore or get rid of from the cell state.

C) It generates the final output y_t .

D) It ignores the current input x_t .

Correct Answer: B

20. In an LSTM, what carries information from initial to later time steps without vanishing easily?

A) The weights

B) The hidden state h_t

C) The cell state C_t

D) The output gate

Correct Answer: C

21. Which activation function is typically used for gates in LSTM and GRU to output values between 0 and 1?

A) Tanh

B) Sigmoid

C) ReLU

D) Softmax

Correct Answer: B

22. What is a key difference between GRU and LSTM?

A) GRU has more gates than LSTM.

B) GRU doesn't contain a separate cell state; it uses hidden states to transport information.

C) LSTM is faster than GRU.

D) GRU cannot be used for sequence modeling.

Correct Answer: B

23. What are the two gates in a GRU?

A) Input and Output

B) Forget and Update

C) Reset and Update

D) Reset and Forget

Correct Answer: C

24. In a GRU, which gate is a combination of the Forget and Input gates?

A) Reset gate

B) Update gate

C) Output gate

D) Hidden gate

Correct Answer: B

25. What is the purpose of the "Reset Gate" in a GRU?

A) To reset the learning rate.

- B) To determine how much past information should be forgotten to avoid gradient explosion.
- C) To output the final prediction.
- D) To add noise to the input.

Correct Answer: B

26. Which architecture "encodes" an input sequence into a single vector and then "decodes" it into an output sequence?

- A) One-to-many RNN
- B) Many-to-one RNN
- C) Sequence to Sequence (Seq2Seq)
- D) Vanilla RNN

Correct Answer: C

27. RNNs are considered "short-term memory" when gradients vanish. How do LSTMs fix this?

- A) By increasing the depth of the network.
- B) By using additive interactions and cell states to improve gradient flow.
- C) By removing all non-linearities.
- D) By using only one gate.

Correct Answer: B

28. In backpropagation through time for an LSTM, what interaction allows for uninterrupted gradient flow?

- A) Matrix multiplication by large weights.
- B) Elementwise multiplication by the forget gate activations.
- C) Squashing the output with tanh.
- D) Adding Gaussian noise.

Correct Answer: B

29. Which SOTA models typically follow LSTM or GRU networks for sequence prediction?

- A) Almost every model based on RNN.
- B) Only models for static images.
- C) Only models without hidden states.
- D) None of the above.

Correct Answer: A

30. What is the result of using a sigmoid in an LSTM gate if the value is closer to 1?

- A) Information should be ignored.
- B) Information should proceed forward.
- C) The gradient should be clipped.
- D) The network should be reset.

Correct Answer: B

Autoencoders

31. What is an Autoencoder?

- A) A supervised learning network for labelling images.
- B) A neural network trained to copy its input to its output using unsupervised learning.
- C) A network used only for reinforcement learning.
- D) An encoder that only compresses data without a decoder.

Correct Answer: B

32. What are the two main parts of an Autoencoder?

- A) Input and Bias
- B) Encoder and Decoder
- C) Hidden layer and Sigmoid
- D) Forward and Backward pass

Correct Answer: B

33. The output of an autoencoder is often called a:

- A) Classification
- B) Prediction
- C) Reconstruction
- D) Probability

Correct Answer: C

34. What does an Autoencoder learn in its hidden layer?

- A) A random noise vector.
- B) A compressed representation or embedding.

- C) The labels for the training data.
- D) The exact identity function for all possible inputs.

Correct Answer: B

35. Why is learning $g(f(x)) = x$ everywhere not useful for an autoencoder?

- A) Because it makes the network too fast.
- B) Because the goal is to learn useful properties of the data, not just an identity function.
- C) Because it requires too much memory.
- D) Because it is impossible to implement.

Correct Answer: B

36. An embedding in an autoencoder is typically:

- A) Higher-dimensional than the input.
- B) A low-dimensional vector on a manifold.
- C) Always a binary string.
- D) Only used during the testing phase.

Correct Answer: B

37. What is an "Undercomplete Autoencoder"?

- A) An autoencoder that is missing its decoder.
- B) An autoencoder where the hidden layer has a higher dimension than the input.
- C) An autoencoder where the hidden code h has a lower dimension than the input x .
- D) An autoencoder that hasn't finished training.

Correct Answer: C

38. An undercomplete autoencoder with a linear decoder and MSE loss is equivalent to:

- A) Support Vector Machines
- B) Principal Component Analysis (PCA)
- C) Logistic Regression
- D) K-Means Clustering

Correct Answer: B

39. What happens if an autoencoder's encoder and decoder have too much capacity?

- A) It learns the data distribution better.
- B) it fails to learn useful features and simply performs a trivial copying task.
- C) It automatically becomes a regularized autoencoder.
- D) It cannot be trained using backpropagation.

Correct Answer: B

40. Which case of autoencoder is designed where the hidden code dimension is GREATER than the input?

- A) Undercomplete
- B) Overcomplete
- C) Symmetrical
- D) Linear

Correct Answer: B

41. How do regularized autoencoders learn useful features even if they are overcomplete?

- A) By increasing the learning rate.
- B) By using a loss function that encourages properties beyond just copying (e.g., sparsity).
- C) By removing the hidden layer.
- D) By using only linear activations.

Correct Answer: B

42. A Sparse Autoencoder includes what in its training criterion?

- A) A noise term on the input.
- B) A sparsity penalty $\Omega(h)$ on the code layer.
- C) A penalty for having too many layers.
- D) An image classification loss.

Correct Answer: B

43. Sparse autoencoders are typically used for which type of task?

- A) Learning features for another task like classification.
- B) Only for exact data replication.
- C) Generating random noise.
- D) Calculating the identity function.

Correct Answer: A

44. Which prior distribution corresponds to an absolute value sparsity penalty?

- A) Gaussian prior
- B) Bernoulli prior
- C) Laplace prior
- D) Uniform prior

Correct Answer: C

45. What is a "Denoising Autoencoder" (DAE)?

- A) An autoencoder that adds noise to its output.
- B) An autoencoder trained to reconstruct clean data from a corrupted input.
- C) An autoencoder that removes music from audio.
- D) An autoencoder that uses no decoder.

Correct Answer: B

46. What does Denoising training force the encoder and decoder to learn?

- A) The specific noise pattern only.
- B) The identity function.
- C) The structure of the data distribution $p_{\text{data}}(\mathbf{x})$.
- D) How to set 50% of nodes to zero permanently.

Correct Answer: C

47. In a DAE, how is the input typically corrupted?

- A) By changing the labels.
- B) By setting a percentage (e.g., 30-50%) of random input nodes to zero.
- C) By doubling the values of all nodes.
- D) By reversing the sequence of pixels.

Correct Answer: B

48. What is a "Contractive Autoencoder" (CAE)?

- A) An autoencoder that contracts its weights to zero.
- B) An autoencoder that penalizes the derivatives of the representation to ensure robustness to small input changes.
- C) An autoencoder that only uses the first half of the data.
- D) A linear autoencoder used for PCA.

Correct Answer: B

49. According to the slides, what is a "Stochastic Autoencoder"?

- A) An autoencoder where the encoder/decoder involves a distribution rather than a simple function.
- B) An autoencoder that uses stochastic gradient descent only.
- C) An autoencoder that works on random data.
- D) An autoencoder that is missing the hidden layer.

Correct Answer: A

50. In a stochastic decoder, we train by minimizing:

- A) The sum of weights.
- B) Negative log-likelihood $-\log(p_{\text{decoder}}(x|h))$.
- C) The variance of the input.
- D) The number of neurons.

Correct Answer: B

51. Which type of autoencoder training method is more "biologically plausible" but rarely used?

- A) Backpropagation
- B) Recirculation
- C) Stochastic gradient descent
- D) Momentum

Correct Answer: B

52. Deep autoencoders offer advantages over single-layer ones because they:

- A) Always require fewer units.
- B) May generalize better and use fewer units overall than a very wide single-layer net.
- C) Do not require backpropagation.
- D) Only work for small datasets.

Correct Answer: B

53. What is a common strategy for training deep autoencoders?

- A) Randomly dropping layers.
- B) Greedily pretraining a stack of shallow autoencoders.

C) Using only linear activations for all layers.

D) Training without a reconstruction loss.

Correct Answer: B

54. What is the theoretical connection of autoencoders in the forefront of generative models?

A) They are exactly the same as CNNs.

B) Their connection to latent variable models (e.g., Variational Autoencoders).

C) They don't use layers.

D) They cannot be generative.

Correct Answer: B

55. Which generative modeling approach is a descendant of the Helmholtz machine?

A) Vanilla RNN

B) ResNet

C) Variational Autoencoder (VAE)

D) LeNet

Correct Answer: C

56. Score matching in autoencoders fits what to the true underlying density?

A) The average value.

B) The slope (score) of the model density.

C) The number of peaks.

D) The weight matrix.

Correct Answer: B

57. In DAE learning, what does each arrow in the learned vector field point towards?

A) Lower probability regions.

B) Higher probability regions.

C) The origin (zero).

D) Random directions.

Correct Answer: B

58. When is the vector field's arrow magnitude near zero in a DAE?

- A) Where the probability is minimum.
- B) Where the probability is maximum (at the data points).
- C) When the weights are large.
- D) When noise is zero.

Correct Answer: B

59. Which loss function is minimized for an autoencoder with a conditionally Gaussian stochastic decoder?

- A) Cross-entropy
- B) Mean Squared Error (MSE).
- C) Hinge loss
- D) Absolute error

Correct Answer: B

60. What type of output distribution corresponds to a softmax in a stochastic decoder?

- A) Gaussian
- B) Bernoulli
- C) Discrete x values.
- D) Laplace

Correct Answer: C

Applications of Autoencoders

61. What was one of the early motivations for studying autoencoders?

- A) Building better GPUs.
- B) Dimensionality reduction.
- C) Only for speech synthesis.
- D) Playing video games.

Correct Answer: B

62. Lower-dimensional representations aid generalization by:

- A) Adding noise to the data.
- B) Providing hints via the mapping to the lower-dimensional space.
- C) Increasing memory consumption.
- D) Making the training slower.

Correct Answer: B

63. Which task benefits from dimensionality reduction by finding entries in a database that resemble a query?

- A) Regression
- B) Information Retrieval.
- C) Sorting
- D) Garbage collection

Correct Answer: B

64. What is the technique called when dimensionality reduction and binarization are used for efficient retrieval?

- A) Linear Probing
- B) Semantic Hashing.
- C) Binary Search
- D) Random Forest

Correct Answer: B

65. How can search be performed efficiently in semantic hashing?

- A) By checking every entry.
- B) By using a hash table and flipping individual bits to find similar entries.
- C) By sorting the database alphabetically.
- D) By recalculating the weights for every query.

Correct Answer: B

66. Semantic hashing has been applied to which types of input?

- A) Only text.
- B) Text and images.
- C) Only audio.
- D) Only integers.

Correct Answer: B

67. To produce binary codes for semantic hashing, which activation function is typically used on the final layer?

- A) ReLU
- B) Sigmoid.

- C) Tanh
- D) Linear

Correct Answer: B

68. What training "trick" can help sigmoid units saturate to nearly 0 or 1 for semantic hashing?

- A) Injecting additive noise just before the sigmoid.
- B) Removing the bias.
- C) Using a very large batch size.
- D) Stopping training early.

Correct Answer: A

69. In the semantic hashing trick, the magnitude of the noise should:

- A) Decrease over time.
- B) Increase over time.
- C) Stay constant.
- D) Be set to zero immediately.

Correct Answer: B

70. The idea of training representations for hashing can be linked to optimizing a loss for:

- A) Maximum weight values.
- B) Finding nearby examples in the hash table.
- C) Minimum number of bits.
- D) Fastest training time.

Correct Answer: B

71. Dimensionality reduction places semantically related examples:

- A) Far away from each other.
- B) Near each other in the lower-dimensional space.
- C) In random order.
- D) only if they have the same label.

Correct Answer: B

72. Which of the following consumes less memory and runtime?

- A) Models of larger spaces.

B) Models of smaller spaces (low-dimensional).

C) Models with no weights.

D) Non-linear models only.

Correct Answer: B

Descriptive Questions

1. Mention the challenges that occur in training a RNN. Explain the solutions to the challenges.
2. Describe the LSTM network architecture with the block diagram.
3. Mention and explain four types of Autoencoders.
4. Explain the Denoising Autoencoder with a computational graph. Describe the DAE training procedure.